

Data File Handling

header file required

`<fstream.h>`

which includes
`iostream.h`

it includes 3 pre defined classes

i) `ifstream` class → used for reading operation (data) from the file.

file to system

- `get()` → for reading character by character.
- `getline()` → to take data line by line
- `read()` → to read complete records (structures)
- `open()` → to open a file
- `close()` → to close a file.
- `seekg()` → to search the element (reading mode)
- `tellg()`

↓
It tells the current position of the point

+ front
- back

eg: i) `seekg(-10, ios::endl);`
↳ from back 10 spaces back.

eg: `ifstream ob;`
`int x = ob.tellg();`

iii) `seekg(10, ios::beg);`

iv) `seekg(10, ios::cur);`

ii) `ofstream` class (output) predefined class

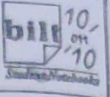
System to file

It is used to write the data in the file.

- `put()` → to write data character wise.
- `write()` → when we want to write record into the file.

3-steps

open a file
perform operation
closing of file

Date	/ /	
Page		

open() →

close() →

seek() → to put the pointer in write place. for

tell() → tells the current position of the pointer. writing mode.

(3) fstream class: (It is derived class of of stream and ifstream)

Open a data file

(i) Using a constructor:

```
ifstream file ("stud.dat");
```

↓
function with parameter (physical file name)
will be called.

→ constructor will be called with parameter.

(ii) Using open() function

```
ifstream file1;  
file1.open ("stud.dat")
```

during opening file will be linked.

Closing of file

file object . close ();

eg: file1 . close ();
↓ no physical file will not be used.
object1

Ques WAP to create a data file text.dat and write 3 lines into it. Read the data from the file and display it.

```
#include <fstream.h>
```

```
#include <conio.h>
```

```
void main ()
```

```
{ char line [80];
```

```
int i;
```

```
ofstream
```

```
file1;
```

```
file1.open ("text.dat");
```

```
file1 << "DOON PUBLIC SCHOOL \n";
```

```
file1 << "B-2 Paschim Vihar";
```

```
file1 << "New Delhi";
```

```
file1.close ();
```

```
cout << "\n Data of file is : \n";
```

```
ifstream
```

```
file2.open ("text.dat");
```

```
for (i=1; i <= 3; i++)
```

```
{ file2.getline (line, 80);
```

```
cout << "\n" << line;
```

```
}
```

```
file2.close ();
```

```
getch();
```

```
}
```

→ file is linked to physical file.
data will be transfer to file1 then to text.dat

as it is attached to file1.

→ reading operation

Ques WAP to create a data file text.dat and write few lines text into it according to requirement. Read the data and display it.

```
#include <stdio.h>
#include <fstream.h>
#include <conio.h>
void main()
{
    char line[80], ch; // for choice
    int i;
    clrscr();
    ofstream file1;
    file1.open("text.dat");
    do
    {
        gets(line);
        file1 << line << "\n";
        cout << "\n Do you want to add more line (y/n)";
        cin >> ch;
    } while (ch == 'y' || ch == 'Y');
    file1.close();
    cout << "\n Data of file is : \n";
    ifstream file2;
    file2.open("text.dat");
    while (!file2.eof())
    {
        file2.getline(line, 80); // end of file
        cout << "\n" << line;
    }
    file2.close();
    getch();
}
```


I LINE BY LINE READING

Date / /
Page

10/07/10
Student Notebook

Ques WAF to read a function data from the file text.dat, count and display no. of line exists in file.

```
void count ( )
{
    char line [80];
    int c = 0;
    ifstream file2;
    file2.open ("text.dat");
    while (!file2.eof())
    {
        file2.getline (line, 80);
        c++;
    }
    file2.close();
    cout << "\n No. of lines" << c;
}
```

Ques WAF to read data from "text.dat", count and display number of lines starting with 'A'.

```
void count ( )
{
    char line [80];
    int c = 0;
    ifstream file1;
    file1.open ("text.dat");
    while (!file1.eof())
    {
        file1.getline (line, 80);
        if (line[0] == 'A')
            c++;
    }
    file1.close();
    cout << "\n No. of lines" << c;
}
```

(mind parameters)
← if we put '.', then it will go to next line.

starting with 'A'

When display line by line

Same as previous.

cout << "\n" << line; (in place of if (line[0] == 'A')
same C++)

Q WAF To read data from "text.dat" and display page-wise (25 lines per page)

```
void display()
{
    char line[80];
    int c = 0;
    ifstream file1;
    file1.open("text.dat");
    while (!file1.eof())
    {
        file1.getline(line, 80);
        cout << "\n" << line;
        c++;
        if (c == 25)
        {
            getch();
            clrscr();
            c = 0;
        }
    }
    file1.close();
}
```


Ques WAF to read the data from the file text1.dat and write into text2.dat after converting into capital alphabets.

```
void convert ()
{
    char line [80];
    ifstream file1;
    file1.open ("text1.dat");
    ofstream file2;
    file2.open ("text2.dat");
    while (!file1.eof())
    {
        file1.getline (line, 80);
        strupr (line);
        file2 << line << "\n";
    }
    file1.close ();
    file2.close ();
}
```

<string.h>

strupr → to
convert whole
string to
Capital

/* → to convert to
small

strlower */

Q WAF to read the data from file text1.dat and write reverse into text2.dat.

```
void convert ()
{
    char line [80];
    if stream file1;
    file1.open ("text1.dat");
    of stream file2;
    file2.open ("text2.dat");
    {
        strrev (line);
        ;
        ;
    }
}
```

if member funcⁿ
not use this

1st file → reading mode
2nd file → writing mode

for (i = 0; i < j; i++, j--)

then use temp to
swap

II. character by character

WAF to read data from "text.dat", count and display no. of vowels.

```
void count()
{
    char ch;
    int c = 0;
    ifstream file1;
    file1.open("text1.dat");
    while (!file1.eof())
    {
        file1.get(ch);
        if (ch == 'A' || ch == 'a' || ch == 'E' || ch == 'e' ||
            ch == 'U' || ch == 'u')
            c++;
    }
    file1.close();
    cout << "No. of vowels = " << c;
}
```

instead of if → switch (ch)

```
case 'A':
    'a':
```

```
case 'U': c++;
}
```

no default.

```
}
file1.close
```


Q WAF to count the no. of blanks.

```
void count()
{
    char ch;
    int c = 0;
    ifstream file1;
    file1.open("text.dat");
    while (!file1.eof())
    {
        if (ch == ' ')
            c++;
        file1.get(ch);
    }
    file1.close();
    cout << "In No. of blanks = " << c;
    for No. of words →  $c = c + 1$ 
    cout << "In No. of words = " << c;

```

```
void count()
{
    char ch;
    int c = 0;
    :
    if (ch == ' ')
        c++;
    c = c + 1;
    cout << c + 1;
}

```

III Word by Word

Q WAF to read data from "text.dat". Count and display word "the" exist no. of times.

```
void count
{
    char word[20];
    int c = 0;
    ifstream file1;

```

* this → special operator,
predefined.

Analogy
to cin
cin >> name;

```
file1.open("text.dat");  
while (! file.eof())
```

```
{  
    if (strcmp(word, "the") == 0)  
        break;  
    file1.getline(word, 20, '\n');
```

```
file1.close();
```

data will be taken
from physical
file and will be
terminated when
blank will come.

condition
true

```
file1.getline(word, 20, '\n');
```

take from file and store in array

fstream class

used for reading as well as for writing.
input, output

```
file1.open("text.dat", ios::in); → reading  
file1.open("text.dat", ios::out); → for writing
```

* In writing case, file does not exist → new file created
* If exists → then it will be erase and pointer will
come in beginning.

* If we want to add data in previous file
then "app" → pointer will come in last.

```
file1.open("text.dat", ios::app);
```

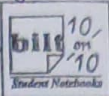
↳ append.

(to add data in
previous file).

Don't prompt] C: > DIR /s; more → list of files alphabetically (page wise)
 C: > DIR /s; more → list of files (page wise)

Date / /

Page



File may be created in modes:

(i) Text file

file1.open("text.dat", ios::in)
 created

- By default file will be in text mode.
- In this case data will be stored and other characters will also be transferred to the file.
- File length always greater than data.
 eg: Amit ↵ (Amit and new line both transferred)

(iv) Binary file

file1.open("text.dat", ios::in, ios::binary).

- In this we have to include ios::binary
- only data is transferred to the file, no other characters will be transferred.
- file length less than text file

eg: Amit ↵ (Only amit will be transferred)
 character

Text file

In this data, as well as some other translations takes place

★ If any modifications:

ios::in, ios::out
 or
 ios::ate

Overlapping

eg

1	3	5	7
---	---	---	---

if we want to write in place of 3

then take pointer one position back and then it will be insertion in place of 3

* `file1.seekg (50, ios::beg)`

from beg pointer will shift 50 bytes forward
→ reading mode.

* `file1.seekg (-50, ios::end);`
from end 50 bytes backward

* `file1.seekg (20, ios::cur);`

20 bytes forward from current position.

Read () function

It is used to read a record from the file when we are doing the work using structure or class.

Syntax :

`file object.read ((char *) &Variable, size of object (class or structure));`

* char pointer

* data from file is stored in object & variable.
↓ class ↓ structure

Write () function

It is used to write a record in the file when we are using structure or class.
* doing the work

Syntax:

file object . write (Char *) & Variable, size of (class) ;
 ↑
 object st structure

first take record from user in variable (if structure) and object (if class)

Ques

WAP to create a structure student containing fields RNo and Name and perform following 7 operations on the file "Stud.dat"

1. Create (if file exists remove all records and add new if does not exist create it)
2. Append
3. Display
4. RNo-wise search
5. Name-wise search
6. Modify
7. Deletion

```
#include <fstream.h>
#include <conio.h>
#include <stdio.h>
#include <string.h>

struct stud
{
    int RNo;
    char name [10];
};
```

```
① void create ()
{
    stud v1; (variable)
    char ch;
    fstream file1;
    file1.open ("stud.dat", ios::out, ios::binary);
    do
```

```

    cout << "\n Enter Roll NO";
    cin >> VI.RNO;
    cout << "\n Enter Name";
    gets (VI.Name);
    file1.write ((char *) &VI, size of (stud));
    cout << "\n Do you want to Add more record (Y/N)";
    cin >> ch;
    } while (ch == 'Y' || ch == 'y');
    file1.close();

```

{data will be transferred from VI to file of size 2+10 = 12 bytes}

② void append()

```

{ stud VI;
  char ch;
  fstream file1;
  file1.open ("stud.dat", ios::app, ios::binary);
  :
  same.

```

③ void Display()

```

{ stud VI;
  fstream file1;
  file1.open ("stud.dat", ios::in, ios::binary);
  while (!file1.eof())
  { file1.read ((char *) &VI, size of (stud));
    cout << "\n Roll NO" << VI.RNO;
    cout << "\n Name" << VI.Name;
  }
  file1.close();
}

```


④ Void Search1 () // Roll No-wise search

```

{stud VI;
  int R, flag = 0;
  fstream file1;
  file1.open ("stud.dat", ios::in | ios::binary);
  cout << "\n Enter Roll NO to be searched";
  cin >> R;
  while (!file1.eof())
  { file1.read ((char *) &VI, size of (stud));
    if (VI.RNO == R)
    { flag = 1;
      cout << "\n Roll NO" << VI.RNO;
      cout << "\n Name" << VI.Name;
    }
  }
  file1.close ();
  if (flag == 0)
  cout << "\n Record not found";
}

```

⑤ Void Search2 () // Enter Name wise

```

{stud VI;
  int flag = 0; char RName[10];
  fstream file1;
  file1.open ("stud.dat", ios::in | ios::binary);
  cout << "\n Enter Name to be searched";
  gets (RName);
  while (!file1.eof())
  { file1.read ((char *) &VI, size of (stud));
    if (strcmp (VI.Name, RName) == 0)
    { flag = 1;
      cout << "\n Roll NO" << VI.RNO;
    }
  }
}

```

```

    cout << "\n Name << V1. Name ;
    file 1. close ( ) ;
    if (flag == 0)
    cout << "\n Record not found";
    }

```

```

(6) void Modify ( )
{
    stud V1;
    int flag = 0, R ;
    ofstream file1, file2;
    file1.open ("stud.dat", ios::in;
                ios::binary);

```

flag = 0

	stud.dat	temp.dat
	1	1
	3	3
	5	6
exists flag = 1	7	7
	9	9

file 2
writing mode

```

file2.open ("temp.dat", ios::out;
            ios::binary);

```

if flag = 1
change the name of
file as stud.dat.

```

    cout << "\n Enter Roll NO  
    whose record to be Modify";

```

```

    cin >> R ;
    while ( ! file1. eof ( ) )
    {
        file1.read ((char *) & V1, size of (stud));
        if ( V1. RNO == R )
        {
            flag = 1 ;
            cout << "\n Enter Roll NO";
            cin >> V1. RNO ;
            cout << "\n Enter Name";
            gets (V1. Name);
            file2.write ((char *) & V1, size of (stud));
        }
        else

```

```

    {
        file2.write ((char *) & V1, size of (stud));
    }
    file1.close ( ) ;
    file2.close ( ) ;

```

if not
matches
same
file
copied


```

if (flag == 1)
{
    remove("stud.dat");
    rename("temp.dat", "stud.dat");
}
else cout << "\n Record not found";
}

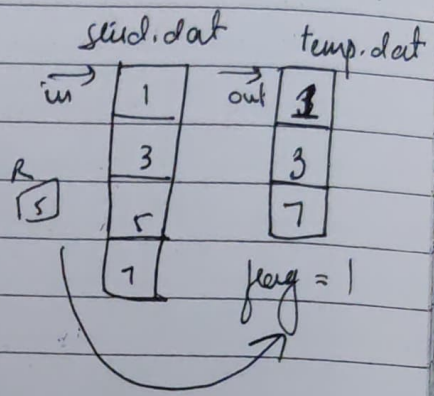
```

(7)

```

void del. ( )
{
    stud V1;
    int flag = 0, R;
    fstream file1, file2;
    file1.open("stud.dat", ios
        :: in, ios :: binary);
    file2.open("temp.dat", ios :: out,
        ios :: binary);
    cout << "\n Enter Roll No whose record
        to be Modified delete ";

```



Delete → del
as → is a keyword

```

    cin >> R;
    while (!file1.eof())
    {
        file1.read((char*) &V1, size of (stud));
        if (V1.RNO == R)
        {
            flag = 1;
        }
        else file2.write((char*) &V1, size (stud));
    }
    file1.close();
    file2.close();
    if (flag == 1)
    {
        remove("stud.dat");
        rename("temp.dat");
    }
    else cout << " Record not found";
}

```

```

}
void main()
{
    int ch;
    do
    {
        clrscr();
        cout << "\n 1 - Create ";
        cout << "\n 2 - Append ";
        cout << "\n 3 - Display ";
        cout << "\n 4 - RNO wise search ";
        cout << "\n 5 - Name wise search ";
        cout << "\n 6 - Modify ";
        cout << "\n 7 - Delete ";
        cout << "\n Enter choice ";
        cin >> ch;
        switch (ch)
        {
            case 1: Create(); break;
            case 2: Append(); break;
            case 3: Display(); break;
            case 4: search1(); break;
            case 5: search2(); break;
            case 6: Modify(); break;
            case 7: Del(); break;
            default: cout << "\n Wrong choice ";
        }
        getch();
    } while (ch >= 1 && ch <= 7);
}

```


WAP to create a class stud (RNO, Name, Enter(), show(), RNO(), RName()) and perform following function on "stud.dat"
Create(), Append(), Display(), search1(), search2(), Modify(), Del().

```
#include <fstream.h>
#include <conio.h>
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
```

class stud

```
{ int RNO;
  char Name[10];
```

(Private data members)

public :

```
void Enter()
```

```
{ cout << "\n Enter Roll NO:";
```

```
cin >> RNO;
```

```
cout << "\n Enter name:";
```

```
  gets(Name);
```

```
}
```

```
void show()
```

```
{ cout << "\n Roll NO" << RNO;
```

```
  cout << "\n Name" << Name;
```

```
}
```

```
int RNO()
```

```
{ return RNO; }
```

```
char * RName()
```

```
{ return Name; }
```

```
};
```

```
void Create()
```

```
{ stud ob;
```

we cannot use ob. as it is

in private (data members)

so we use

R.RNO (return type)

for it
returns
address
so pointer

```

char ch;
fstream file1;
file1.open ("stud.dat", ios::out, ios::binary);
do
{
    ob.Enter ();
    file1.write ((char *) &ob, size of (stud));
    cout << "Do you want to add more record (y/n)";
    cin >> ch;
} while (ch == 'Y' || ch == 'y')
file1.close ();
}

```

② void Append ()

```

{
    stud ob;
    char ch;
    fstream file1;
    file1.open ("stud.dat", ios::app, ios::binary);
    do {
        ob.Enter ();
        file1.write ((char *) &ob, size of (stud));
        cout << "Do you want to add more Record (Y/N)";
        cin >> ch;
    } while (ch == 'Y' || ch == 'y');
    file1.close ();
}

```

③ void display ()

```

{
    stud ob;
    fstream file1;
    file1.open ("stud.dat", ios::in, ios::binary);
    while (!file1.eof())
    {
        file1.read ((char *) &ob, size of (stud));
    }
}

```



```

ob. show();
}
file.close();
}

```

(4) void search1() // Roll No. wise

```

{
    stud ob;
    int R, flag = 0;
    fstream file1;
    file1.open("stud.dat", ios::in | ios::binary);
    cout << "\n Enter Roll No to be searched";
    cin >> R;
    while (!file1.eof())
    {
        file1.read((char *) &ob, sizeof(stud));
        if (ob.RNO() == R)
        {
            flag = 1;
            ob.show();
        }
    }
    file1.close();
    if (flag == 0)
        cout << "\n Record not found";
}

```

(5) void search2() // Name wise

```

{
    stud ob;
    int flag = 0;
    fstream file1;
    file1.open("stud.dat", ios::in | ios::binary);
    cout << "\n Enter Name to be searched";
    gets(sname);
    while (!file1.eof())
    {
        file1.read((char *) &ob, sizeof(stud));
    }
}

```

```

if (strcmp (Ob. RName, SName) == 0)
{
    flag = 1;
    Ob. show ();
}
file1. close ();
if (flag == 0)
    cout << "\n Record not found";
}

```

1/2 Void Modify ()

```

{
    stud ob;
    int R, flag = 0;
    fstream file1, file2;
    file1. open ("stud. dat", ios::in, ios::binary);
    file2. open ("temp. dat", ios::out, ios::binary);
    cout << "\n Enter Roll NO. whose record to be modified";
    cin >> R;
    while (!file1. eof ())
    {
        file1. read ((char *)& ob, size of (stud));
        if (ob. RRNO () == R)
        {
            flag = 1;
            ob. show ();
            ob. Enter ();
            file2. write ((char *)& ob, size of (stud));
        }
        else
            file2. write ((char *)& ob, size of (stud));
    }
    file1. close ();
    file2. close ();
    if (flag == 1)
        remove ("stud. dat");
}

```



```

rename ("temp.dat", "stud.dat");
}
else cout << "\n Record not found";
}

```

(7)

```

void del ()
{
    stud ob;
    int R, flag = 0;
    fstream file1, file2;
    file1.open ("stud.dat", ios::in, ios::binary);
    file2.open ("temp.dat", ios::out, ios::binary);
    cout << "\n Enter Roll no whose record to be deleted";
    cin >> R;
    while (!file1.eof())
    {
        file1.read ((char *) & ob, sizeof (stud));
        if (ob.RRNO == R)
        {
            flag = 1;
        }
        else file2.write ((char *) & ob, sizeof (stud));
    }
    file1.close();
    file2.close();
    if (flag == 1)
    {
        remove ("stud.dat");
        rename ("temp.dat", "stud.dat");
    }
    else cout << "\n Record not found";
}

```

Void Main

(Same as structure)


```
File.open("Item.dat", ios::binary | ios::
in::out);
```

```
int CINO.
```

```
cout << "\n Item No to modify price";
```

```
cin >> CINO.
```

```
while (File.read((char *) this, size of (Item)))
```

```
{ if (CINO == INA)
```

```
cout << "\n Present Price" << Price;
```

```
cout << "\n charged Price";
```

```
cin >> Price;
```

```
int FilePos = File.tellg();
```

```
File.seekp (FilePos - size of (Item), ios::beg);
```

```
File.write ((char *) this, size of (Item));
```

```
}
```

```
}
```

```
File.close();
```

```
}
```

Ques

WAP to print the data of a file "text.dat" on printer.

IPRN

```
#include <fstream.h>
void main()
{
    char ch;
    fstream file1, file2;
    file1.open("text.dat", ios::in);
    file2.open("PRN", ios::out);
    while (!file1.eof())
    {
        file1.get(ch);
        file2.put(ch);
    }
    file1.close();
    file2.close();
}
```

→ will not create file, will transfer directly to the printer.

Output

```
void main()
{
    char ch = 'A';
    fstream fileout("data.dat", ios::out);
    fileout << ch;
    int p = fileout.tellp();
    cout << p;
}
```

↑
writing mode

What is the output of the file content before the execution of the program is the string "ABC"

"ABC" "A"

Ques Given a binary file "SPORTS.DAT" of the following structure type

```
struct sports
{
    char Event[20];
    char Participants[10][30];
};
```

Write

what a functions in C++ that would read contents from "SPORTS.DAT" and create a file "Athletic.dat".
Copy only those records where the event name is Athletics.

```
void Copy(.)
{
    sports v1;
    file1.open("SPORTS.DAT", ios::in, ios::binary);
    file2.open("Athletic.dat", ios::out, ios::binary);
    while (!file1.eof())
    {
        file1.read((char *)&v1, size of (sports));
        if (strcmp(Event, "Athletics") == 0)
        {
            file2.write((char *)&v1, size of (sports));
        }
    }
    file1.close();
    file2.close();
}
```